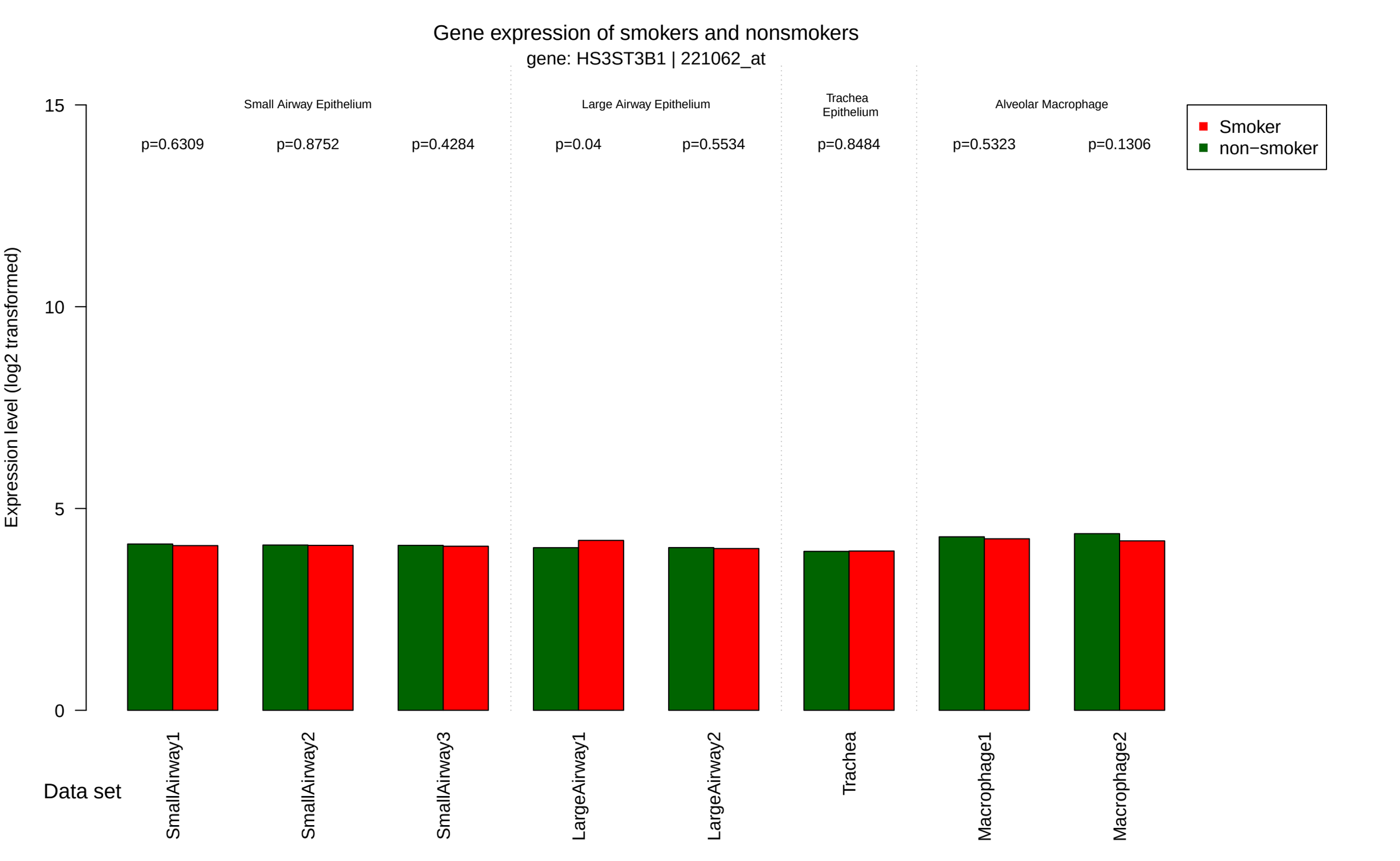
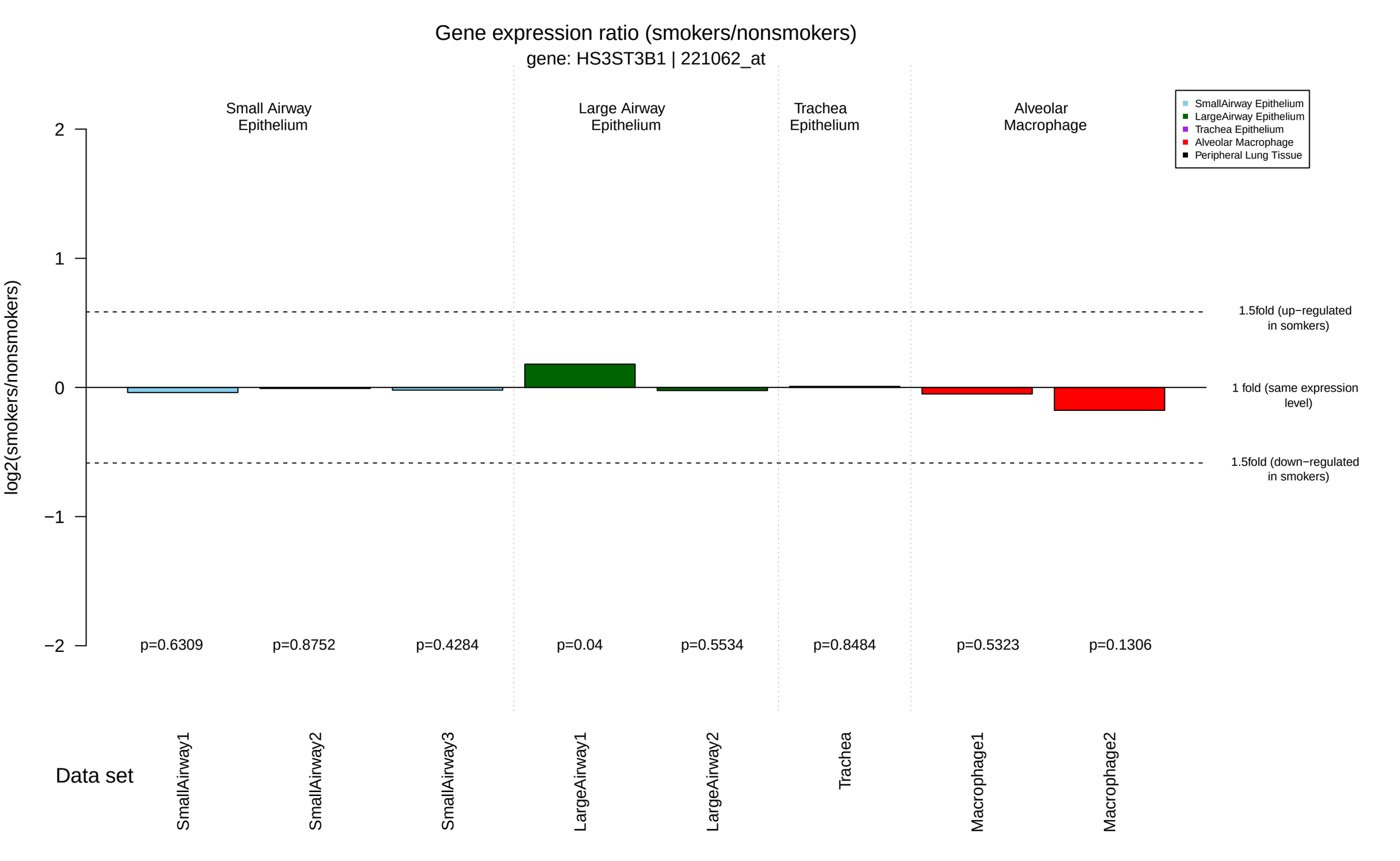
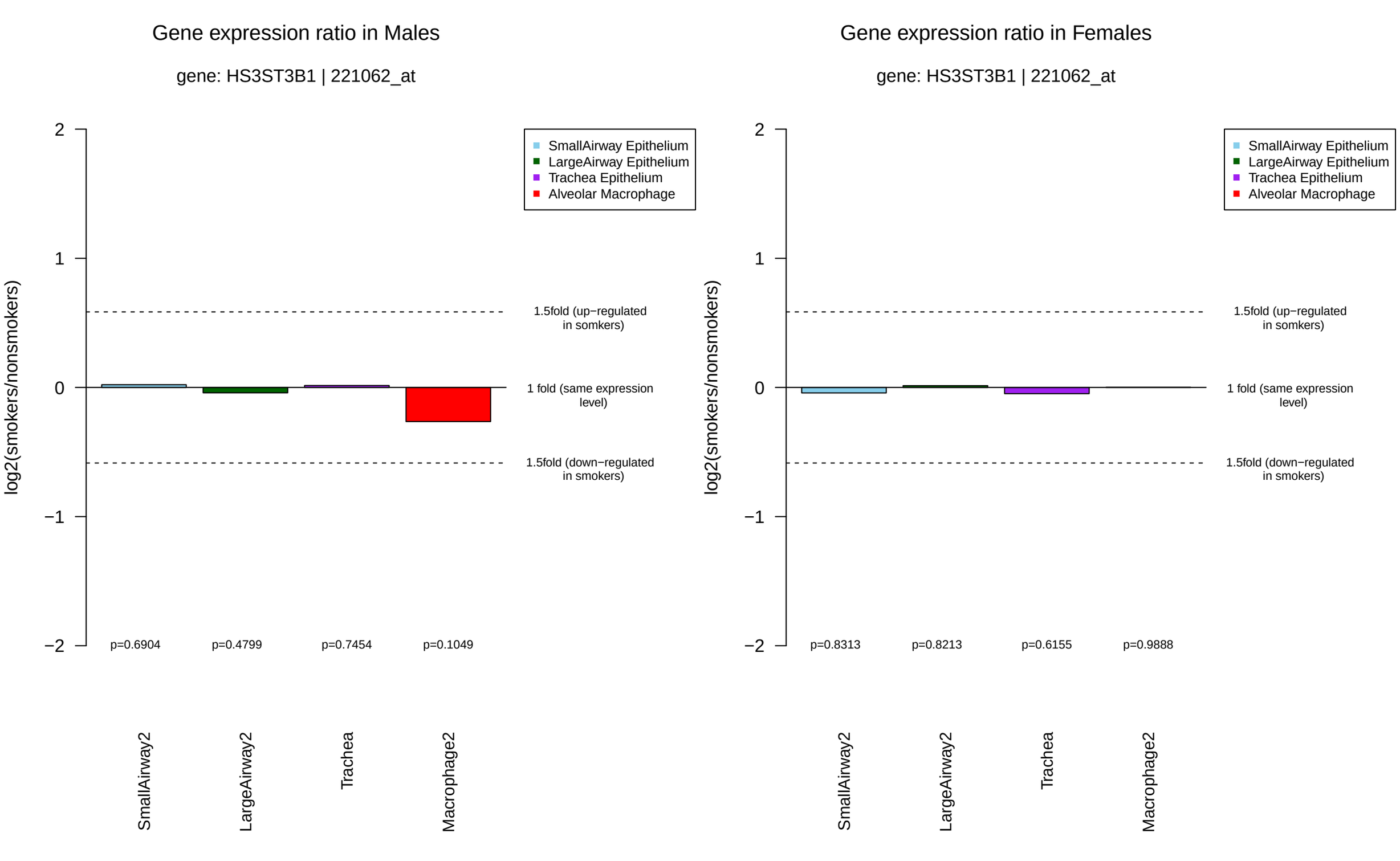
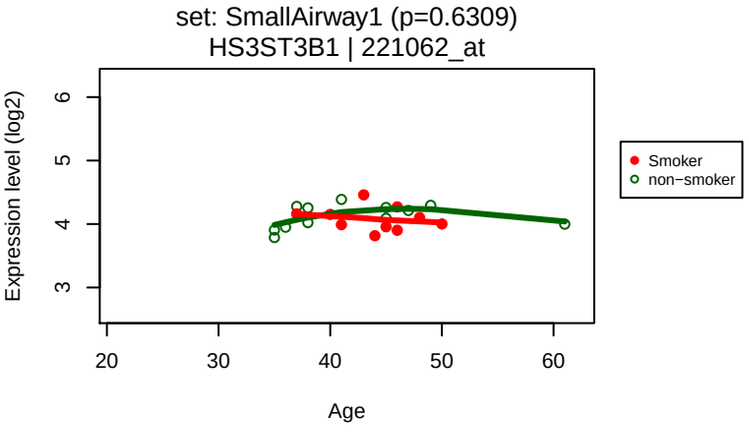
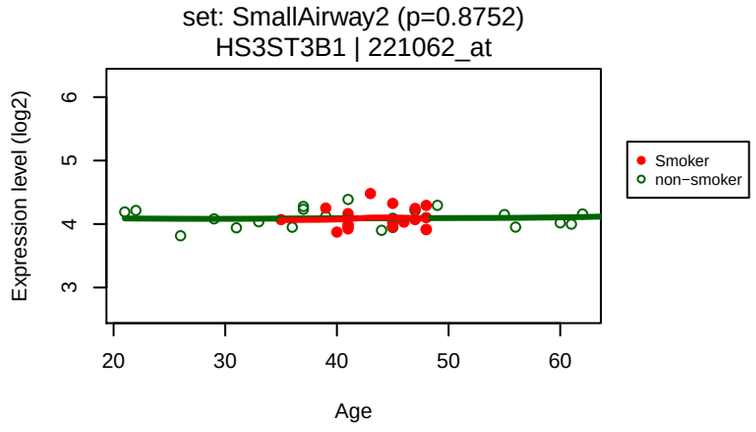
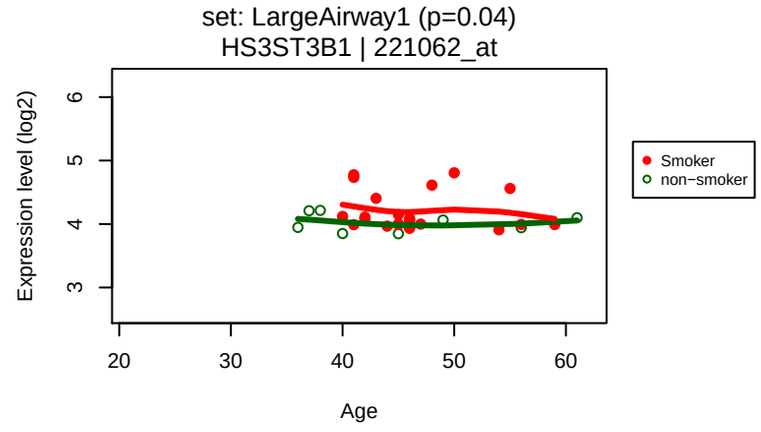

Gene expression of smokers and nonsmokers
gene: HS3ST3B1 | 221062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6309
p=0.8752
p=0.4284
p=0.04
p=0.5534
p=0.8484
p=0.5323
p=0.1306
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

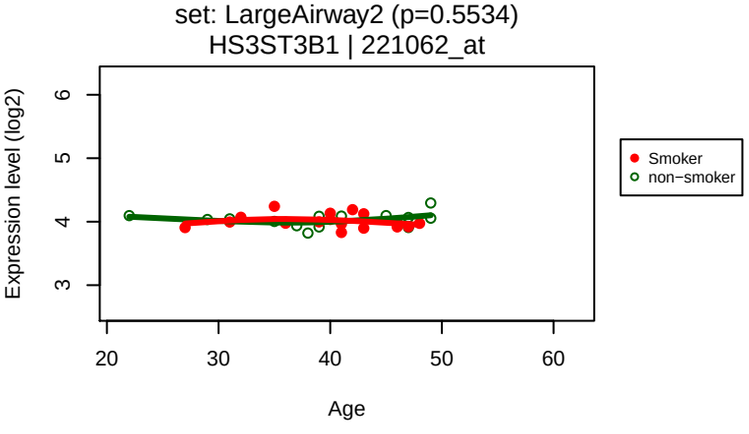

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST3B1 | 221062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6309
p=0.8752
p=0.4284
p=0.04
p=0.5534
p=0.8484
p=0.5323
p=0.1306
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

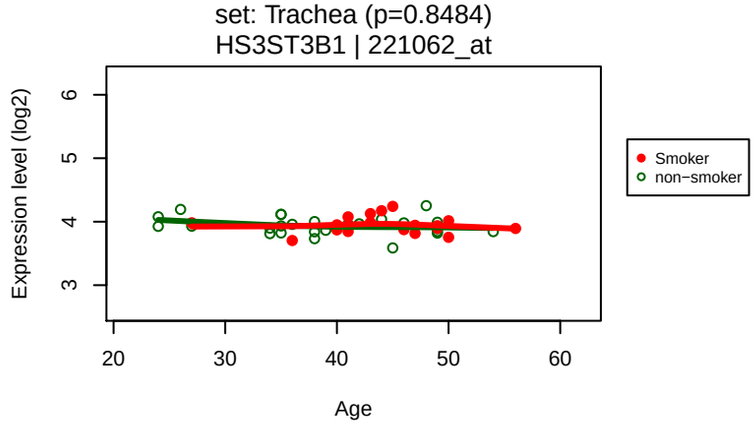

Gene expression ratio in Males
gene: HS3ST3B1 | 221062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6904
p=0.4799
p=0.7454
p=0.1049
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST3B1 | 221062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8313
p=0.8213
p=0.6155
p=0.9888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

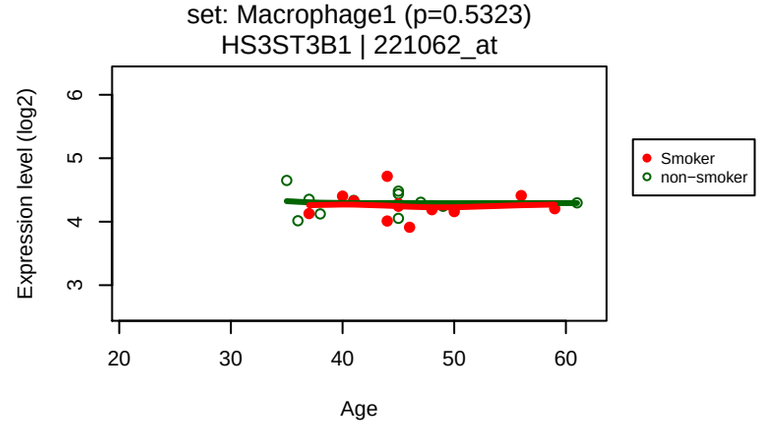

set: SmallAirway1 (p=0.6309)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker

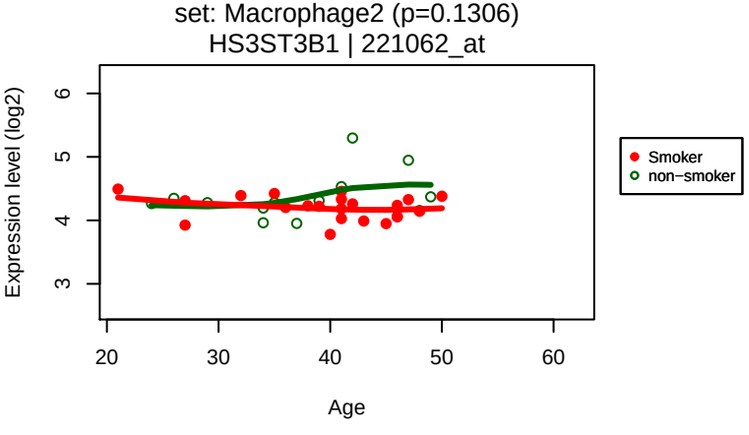

set: SmallAirway2 (p=0.8752)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker

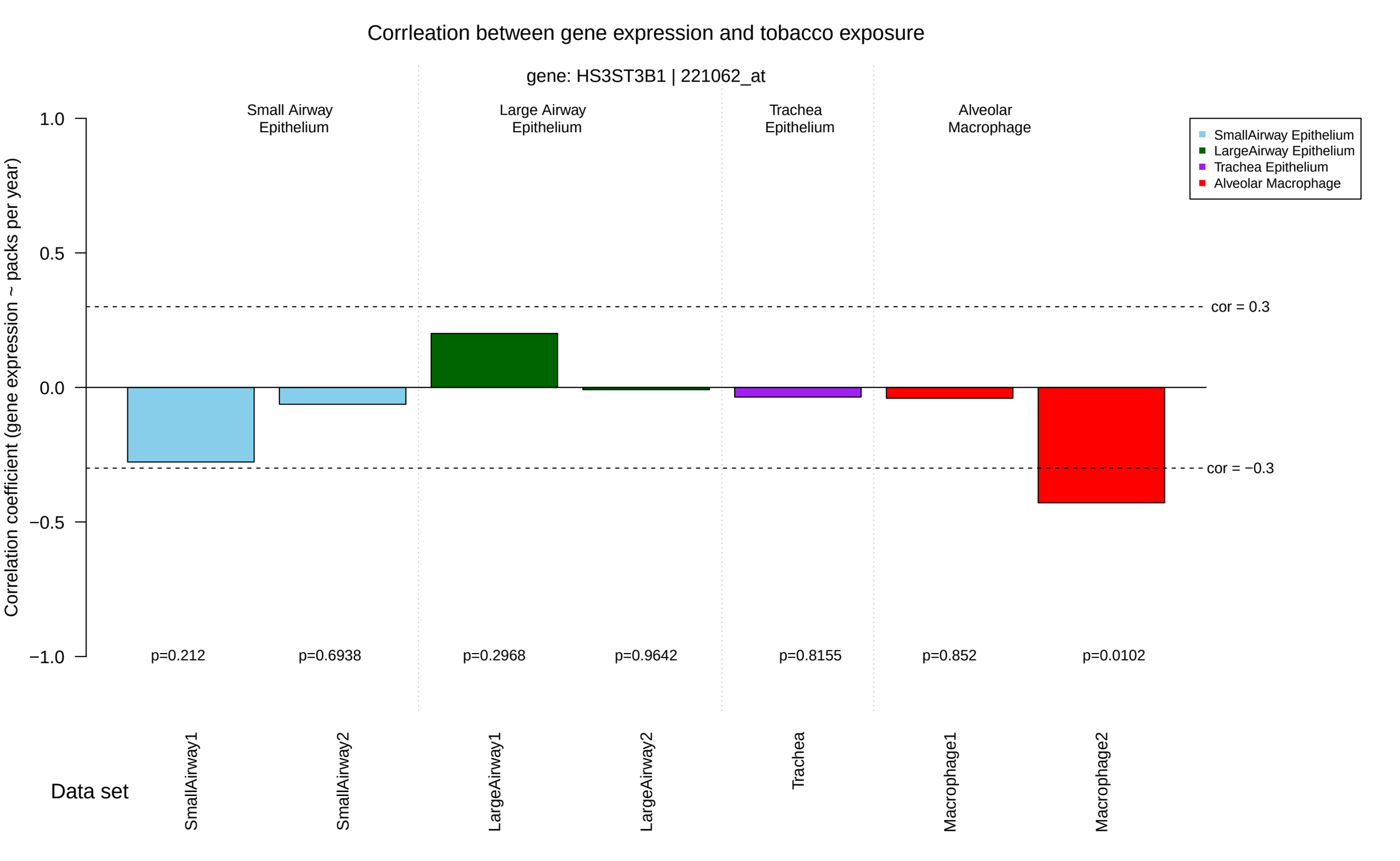

set: LargeAirway1 (p=0.04)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5534)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8484)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5323)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1306)
HS3ST3B1 | 221062_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST3B1 | 221062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.212
p=0.6938
p=0.2968
p=0.9642
p=0.8155
p=0.852
p=0.0102
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2